Why Separable Subjects Still Fail to Link: A Streaming Journal over Knesset Committee Protocols

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Abstract

We build a longitudinal journal over a stream of Israeli Knesset Finance Committee transcripts, coupling three tasks — topic segmentation, *streaming* cross-document subject linking, and update summarization — under an asymmetric cost model in which merging two distinct matters is worse than splitting one. Our central result is negative and, we argue, the most useful thing the project produced. On a manually annotated set of 523 segments, one fixed representation (`multilingual-e5-large`+ABTT) scores F_1=0.619 judging subject identity over *pairs*, and F_1=0.125 when the *same representation* makes the *same decision* online against a growing journal. The collapse is caused not by weak Hebrew embeddings but by the streaming decision structure, where only 6.9% of arrivals are true repeats. Under a matched streaming-honest protocol a distinctiveness *margin* rule raises F_1 0.087→0.190, while every LLM verifier stays ≤ 0.111. Two annotators rating all 22 gold journal chains find them coherent (4.16/5) but decreasingly faithful the longer a matter recurs (3.91→1.75), a degradation our LLM judge — which never scores a chain below 60/100 — does not detect at all. We further document two ways this benchmark rewards look-ahead.

1 Introduction

A parliamentary committee returns to the same matter — a bill, a budget transfer, a programme — again and again over months, so reconstructing the history of any one matter means reading dozens of unrelated transcripts. We produce that history automatically: an index in which each entry is one underlying matter carrying a dated log of its progress. The problem decomposes into three coupled tasks. **Segmentation** partitions each protocol into subject spans; **streaming subject linking** decides, for each arriving span, whether it continues an existing entry or opens a new one; and **log writing** generates only the *incremental* update to the matched entry.

How this is done today, and its limits. Task 2 is a variant of cross-document event coreference (Cybulska and Vossen, 2014; Barhom et al., 2019), but the standard setting is *offline*: a system sees the whole collection and clusters it globally. Ours is online over a growing index, with no look-ahead and its own past errors persisting. Task 3 is update summarization (Dang and Owczarzak, 2008) with retrieval augmentation (Lewis et al., 2020); unlike query-conditioned RAG, faithfulness is defined relative to a *chain of prior summaries* rather than a source document, because the retrieved log records what has already been written.

Asymmetric cost. Merging two distinct matters corrupts an entry permanently and silently; splitting one is visible and repairable. We therefore report every operating point twice: at best F_1, and at the *anti-duplication* point where the false-merge rate (FMR) is held $\leq 5\%$.

Contributions. We expected the bottleneck to be Hebrew representation quality; it is not. Our research question became: *why does near-ceiling subject separability fail to produce usable streaming linking, and what closes the gap?* We contribute (i) a manual gold standard over 211/213 protocols, and an automatic segmenter whose residual error is split-heavy by design; (ii) a quantification of the separability/decision gap with a base-rate explanation; (iii) a matched ablation of the levers that close it, all structural rather than representational; (iv) evidence that an LLM helps as an *extractor* but not a *classifier*; (v) a two-annotator reading of the finished journal that our LLM judge does not predict; and (vi) two look-ahead leaks found in our own pipeline. Any index built incrementally over a stream faces the same arithmetic, and our diagnosis predicts a better encoder will not help.[1]

[1] Code, gold annotations and per-step outputs: `https://github.com/yoel-marcu/knesset-protocol-journal`

Statistic	Value
Protocols (meetings)	213
Protocols manually segmented	211
Protocol-level canonical topics	151
Gold segments (subject spans)	523
Distinct subject labels	487
Recurring matters (≥ 2 appearances)	22
Repeat events (decidable links)	36
Base rate of repeats among arrivals	6.9%

Table 1: Data statistics. Real-world Hebrew parliamentary transcript, 25th Knesset Finance Committee, 2022-11 to 2023-10. *Repeat events* are the only arrivals with a correct link available, and their scarcity drives every result that follows.

2 Data

We use 213 Hebrew transcripts of the 25th Knesset Finance Committee (2022-11-21 to 2023-10-26); each is one meeting, given as committee, ISO date and the full transcript. Protocol-level subject markers yield 151 canonical topics after fuzzy variant merging, but these describe the *meeting*, not the spans inside it, so we manually segmented 211/213 protocols into subject spans labelled with the matter under discussion (Table 1).

Two properties dominate what follows. We deliberately did *not* merge near-duplicate labels across protocols — under our cost rule an unmerged duplicate is a cheap false split, whereas a wrong merge would corrupt the gold itself — so recurrence counts are conservative. And, decisively, repeats are *rare*: 36 of 523 arrivals continue an existing matter, a base rate of 6.9%.

3 Methods

Automatic segmentation. Deployment requires segmenting incoming protocols automatically. An instruction-tuned LLM (`gemma-4-31B-it`, 4-bit NF4, greedy) reads sliding windows of 30 utterances (overlap 6) and emits JSON cuts: *after* a given utterance, or *within* one, anchored by a verbatim quote we verify against the source. The prompt (Fig. 5) encodes committee structure — a speaker change is never a boundary, a vote closes its matter, each budget request (*pniya*) opens one — and instructs the model to cut when uncertain, since over-segmentation is the cheap error under our cost model. Weaker models failed characteristically: `llama-3.1-7b-instruct` produced spans unrelated to the discourse and cut mid-sentence; `dictalm3.0` (7B) collapsed segmentation into diarization, cutting on speaker changes in violation of the prompt's most emphasised rule. The segmenter runs over an extended 927-protocol corpus (2022-11 through 2026-05), of which the 213 protocols of Section 2 are the annotated prefix.

Representation. Segments are embedded with `multilingual-e5-large` (Wang et al., 2024), using discussion content only — no headers, agendas, attendees or speaker names. Dense encoders are severely anisotropic (Ethayarajh, 2019): here cross-topic cosine averages ≈ 0.95, leaving a same/different gap of 0.007–0.066. We apply All-But-The-Top (ABTT) (Mu and Viswanath, 2018) — centre, project out the top-k principal directions, re-normalise — and compare against TF-IDF and AlephBERT (Seker et al., 2022).

Two evaluation protocols. **Pairwise** scores all segment pairs and measures *separability*, with oracle access to the full set. **Streaming** presents segments chronologically and scores each against the journal built so far — the actual task.

Levers. An entry is represented either by its *first span, frozen* or by the running *centroid* of its members; the rule is either a flat threshold ($\cos_{(1)} \geq \theta$) or a *distinctiveness margin* ($\cos_{(1)} - \cos_{(2)} \geq \theta$). We vary one lever at a time, CPU-only, holding transform, fitting protocol and growth model fixed.

LLM as verifier, LLM as extractor. Following retrieve-then-verify we shortlist by dense retrieval and ask an LLM to classify *same*/*related*/*new*, from a candidate snippet and from its *full accumulated timeline*. Separately we use it as an *extractor* of stable fingerprints (ministries, laws, programmes, request numbers), fusing identifier overlap additively with the geometric margin. Models: `dictalm2` (Shmidman et al., 2024) and `qwen2.5` at 3B/7B.

Streaming-honest fitting. ABTT and whitening estimate parameters *from data*; fitting once over the whole matrix uses segments that have not arrived. We refit at every step from the observed prefix, and report batch and honest numbers side by side.

Log writing. Each update is conditioned on the running journal built from the model's *own* prior generations, so faithfulness is chain-relative. Since predicted links are only $\approx 13\%$ precise, evaluating through them would conflate two failures; we use **gold chains** (22 matters, 58 entries), judged by `Qwen2.5-3B-Instruct` — neither generator — for *faithfulness* and *novelty*.

Representation	ARI	AUC
e5, no ABTT (k=0)	0.124	0.857
e5 + ABTT (k=1)	**0.657**	**0.954**
TF-IDF	0.382	0.874
AlephBERT + ABTT (k=2)	0.265	0.854

Table 2: Segment representation on the gold set. ARI is agglomerative clustering against gold subject labels; AUC is same/different-matter separability over segment pairs.

Protocol	AUC	F_1	P	R	FMR
Pairwise (oracle)	0.954	**0.619**	0.500	0.811	0.038
Streaming (real)	—	**0.125**	0.077	0.333	0.296

Table 3: The separability/decision gap. Identical embeddings, identical gold data, identical metric; only the decision setting changes.

Human assessment. The same 22 gold chains — all of them, so there is no sampling decision to defend — were also read end to end by two annotators, independently, and scored 1–5 on two axes: *longitudinal coherence* (does the sequence read as one continuous, non-contradictory matter?) and *faithfulness* (is each update supported by its meeting and genuinely incremental, rather than a restatement?). That is 44 chain-level ratings against the generations Table 7 scores, on the best configuration (`dictalm2`, zero-shot) over gold links, so it measures summarisation and not linking.

4 Results

De-anisotropisation matters more than encoder choice. Table 2 shows that removing a *single* principal direction moves `e5` from ARI 0.124 to 0.657 — more than the entire spread across `e5`, TF-IDF and AlephBERT. Before ABTT, TF-IDF beats every dense model: the discriminative signal is lexical and the dense geometry is dominated by shared bureaucratic register. We fix `e5`+ABTT-k1 throughout.

The separability/decision gap. Table 3 and Fig. 1 give the central result: a representation separating same- from different-matter pairs at F_1 0.619 collapses to F_1 0.125 when the same decision is made online. **The bottleneck is the streaming decision structure, not representation quality.**

The cause is a base-rate effect, plotted in Fig. 2. Only 36 of 523 arrivals are repeats, and each is compared against a journal growing toward several hundred entries, so with n candidates expected false merges scale as $n \cdot \mathrm{FPR}$ while true positives stay at most one. At the threshold that is optimal *pairwise* (θ=0.26) the per-pair false-positive rate is 0.027 — excellent — yet it passes one expected false merge per arrival once the journal holds 38 entries, which happens 52 arrivals in, and ends the stream at 13 expected false merges against 0.056 expected true ones. Notably, **retrieval was never the bottleneck** (Fig. 3): ranking places the correct entry in the top 10 for *all* 36 repeats (recall@1 0.694, @5 0.944, @10 1.000). The information is in the shortlist; the *decision* fails. Table 4(a) shows why: a segment and its true continuation — same ministry, same request number — score 0.266, while an unrelated ministry's transfer scores 0.448, shared administrative phrasing outweighing the sparse identifying tokens.

What closes the gap. Table 5 varies one lever at a time with everything else fixed. The **distinctiveness margin** is the decisive change: F_1 0.087 $\rightarrow$ 0.190 and recall at FMR$\leq$ 5% from 0% to 8.6%. Fig. 4 shows why it works where a threshold cannot: the top-1 cosine of an arrival that should link is not merely uninformative but slightly *lower* than that of an arrival that should not (median 0.421 vs 0.503), while the gap to the runner-up separates them in the right direction (0.138 vs 0.061). With families of near-identical templates, high absolute similarity says only that the journal contains another budget transfer; a large *margin* says this one is specifically about the same matter. It is the same failure mode as Table 4(a).

Replacing the frozen first span with the running centroid is a large *recall* fix: on the batch-fit ABTT-k1 path it lifts recall 32% $\rightarrow$ 89% (F_1 0.103 $\rightarrow$ 0.107), because the streaming setting keeps supplying evidence and the baseline discards it. We report it on that path because under the matched protocol the threshold rule degenerates to accepting almost everything, leaving the two threshold rows indistinguishable. It improves the entry representation without, on its own, converting into F_1.

Extractor, not classifier. As a **verifier** judging from a candidate snippet the best LLM reached F_1 0.111; given the *full timeline* it got *worse* (0.068), the longer context increasing its willingness to merge. As an **extractor** of fingerprints fused with the geometric score it helped: under a common batch-fit setting, geometry alone 0.2000 versus 0.2407 (`dictalm2`) and 0.2013 (`qwen7b`), and a hard identifier gate reaches 11.4% recall at

(a) Why the decision fails: the register confound	*gold repeat event 8; e5+ABTT-$k1$ cosine*
Query 2022-12-13, Energy Ministry	“Ministry of Energy. *Request no. 177* is intended to fund 559 million shekels in commitment authority and 4 staff positions…”
True continuation 2022-12-06 $\cos = \mathbf{0.266}$	“Request approved. *Request no. 177*: Ministry of Energy. The request concerns the Ministry of Energy budget line…”
Top distractor 2022-12-06, Transport $\cos = \mathbf{0.448}$	“…they are very nice people, they smile, and they bring the request again in the same way. Ministry of Transport…”
*The true match shares the ministry **and the request number**; the distractor shares only bureaucratic phrasing, and still outranks it by* 0.18. *This is what the distinctiveness margin and the fingerprint extractor each attack.*	

(b) Why log writing fails: recap-then-append	*chain “rising milk prices”, dictalm2, zero-shot*
Entry 1 2022-11-28 (opening)	“…controlled milk prices have not changed in a decade, but input costs rose, so an agreement was reached with the government on a moderate 4.8% rise. The chair, Moshe Gafni, asked why Israel does not subsidise dairy farmers as in Europe and the US, where there is **no VAT on dairy**, and why **consumers must absorb the increase** instead of the state helping the industry.”
Entry 2 2022-11-28 (*should be the increment*)	“The chair, Moshe Gafni, stressed the need to protect dairy farmers and prevent shifting the burden to consumers. He recalled the difference between state support elsewhere, where **there is no VAT on dairy**, and Israel, where **consumers must absorb the increase**. Gafni added that the dairies must not exploit the situation to raise *unregulated* prices, and that the whole chain must be handled properly.”
Bold marks content already present in entry 1. The genuinely new material is one clause — unregulated prices and the full supply chain — wrapped in a restatement of the previous entry. Novelty scores 18.1/100 while faithfulness scores 80.7: the model is truthful about the record and unable to emit only the diff. Human readers mark this chain down for faithfulness, and mark longer chains down further (Table 8).	

Table 4: Qualitative examples behind the two central failures. Hebrew source text translated by the authors; ellipses mark elision, emphasis added. (a) is the retrieval confound of Section 4; (b) is the novelty failure of Table 7.

Method	$\mathbf{F_1}$	R@FMR5
matched: centering, honest refit, oracle growth		
First span + threshold	0.086	0%
Centroid + threshold	0.087	0%
Centroid + **margin**	**0.190**	**8.6%**
LLM as classifier		
Best verifier (single snippet)	0.111	≈0%
Verifier, full timeline	0.068	0%

Table 5: What closes the gap. **R@FMR5** is recall at the anti-duplication point (FMR$\leq$ 5%). The first block varies *one lever at a time* with the transform, its fitting protocol and the growth model held fixed (`honest_levers.json`); only the margin converts into F_1. Under this protocol the threshold rule degenerates — both threshold rows peak at recall 1.0 with precision ≈0.045 — so the centroid's benefit is not separable here; it is visible on the batch-fit ABTT-$k1$ path, where it lifts recall 32%→89%.

Transform	Batch (leaky)	Honest refit
Whitening (50-dim)	0.250	0.094
Centering	0.175	0.190
ABTT-$k1$	0.200	0.162

Table 6: Look-ahead leak 1: the fitting protocol as an experimental variable (best F_1, streaming linking). Transforms needing many samples cannot be fitted from an early prefix and lose most of their apparent advantage.

	`dictalm2`		`qwen2.5-7b`	
Prompt	**Faith.**	**Nov.**	**Faith.**	**Nov.**
Zero-shot	**80.7**	**18.1**	63.4	**10.1**
+ example (real)	71.1	15.0	63.1	8.3
+ example (fictional)	53.6	5.0	**69.2**	5.3

Table 7: Context-conditioned log writing on 22 gold chains (58 entries), scored 0–100 by `Qwen2.5-3B-Instruct`. Faithfulness is acceptable; novelty is the failure, and few-shot examples make it monotonically worse for both generators.

FMR$\leq$ 5% — four times the baseline. It is the natural remedy for Table 4(a).

Two look-ahead leaks. Table 6 treats the fitting protocol as an experimental variable. **(1) Batch-fit transforms.** Whitening needs a 50-dimensional covariance estimate that does not exist early in the stream; refit honestly it collapses $0.250 \to 0.094$, while cheap transforms survive. **(2) Undefined cold-start margins.** With one entry the runner-up is undefined; a sentinel -1 makes the margin $\cos +1$, clearing any threshold, so the second segment of every run merges unconditionally. **On streaming benchmarks the fitting protocol must be reported alongside the metric.**

Log writing: faithful but not incremental. Table 7 shows faithfulness is acceptable — the Hebrew-native `dictalm2` is clearly more grounded than the stronger general-purpose model — but novelty is the failure. Table 4(b) shows the pat-

22 gold chains, 2 raters	Coher.	Faith.
Rater A (mean ± sd)	3.77 ± 1.45	3.05 ± 1.56
Rater B	4.55 ± 1.01	3.86 ± 1.28
Pooled	**4.16**	**3.45**
Cohen's κ	0.353	0.113
κ, quadratic weights	0.426	0.282
Pooled, 2-entry (n=16)	4.44	3.91
Pooled, 3-entry (n=4)	3.75	2.50
Pooled, 5-/9-entry (n=2)	**2.75**	**1.75**
Spearman ρ of the pooled human score against		
chain length	-0.482^{*}	-0.663^{**}
judge faithfulness	$+0.201$	-0.042
judge novelty	-0.193	-0.208

Table 8: Human assessment of the finished journal: 44 chain-level ratings over the same 22 chains and the same generations Table 7 scores. Both raters see the journal as coherent, both see faithfulness fall as a matter recurs, and neither is predicted by the 3B judge ($^{*}p<0.05$, $^{**}p<0.01$; all judge correlations $p>0.35$). The two raters agree only moderately on faithfulness, which bounds how sharp any of these numbers can be.

tern: the second entry re-establishes the first almost point for point before adding anything. Few-shot prompting made this *monotonically worse* for both models (18.1→15.0→5.0; 10.1→8.3→5.3); in the worst condition 92% of updates were near-verbatim copies of the example's answer. For a free-text diff task, examples are imitated as *content* rather than *style* — the opposite of what we saw for classification.

What humans see that the judge does not. Coherence is high (pooled $4.16/5$): the journal reads as one continuous matter, which is what gold linking should buy. Pooled faithfulness is $3.45/5$ and, decisively, not uniform: it falls from 3.91 on two-entry chains to 2.50 on three-entry and 1.75 on the two long ones ($\rho=-0.663$, $p=0.001$; significant for each rater separately). **The recap-then-append failure of Table 4(b) compounds**: each update restates more of a longer history, so a defect that is mild per entry is what makes long chains unusable — and long chains are the point of a journal.

The judge tracks neither reader: pooled human faithfulness correlates $\rho=-0.042$ with the judge's, and no judge correlation in Table 8 reaches $p<0.35$. The mechanism is its dynamic range — every chain scores between 60 and 95, so it has no vocabulary for a bad chain. The six chains a human scored $1/5$ average 80.6 from the judge, and are not marginal: rater notes record an opposition MK presented as the incoming committee chair, an MK assigned to the wrong party, and an "update" announcing a development that does not

Segmenter vs. LLM-judge reference	Value
Judged protocols / system segments	18 / 356
Boundary F_1 (±0/±1/±2)	0.51 / 0.59 / 0.61
P_k / WindowDiff	0.31 / 0.39
Segment-count ratio (system/judge)	1.58
Confirmed merges (merge-free prot.)	24 (7/18)
Within-anchors verified verbatim	95.9%

Table 9: Blind-first LLM-as-judge assessment of the segmenter, scored against the judge's own reference under the asymmetric cost rule (merges heavy, splits light). The judge segments each protocol before seeing the system output.

exist. They also caught what no automatic metric reported: in 2 of 22 chains the generator emits *identical* update text for two different meetings — recap-then-append at its limit. Human agreement is itself only moderate (κ_{quad} 0.43 and 0.28), so these ratings are a floor on the task's measurability, not a gold standard — but two readers who agree only moderately with each other both find the length effect, and neither finds the judge.

End-to-end journal. Run on all 523 gold segments under predicted growth, the anti-duplication point ($\theta=0.342$) yields 491 entries, 25 multi-session, at 0.950 purity; the best-F_1 point ($\theta=0.156$) yields 343 entries, 80 multi-session, at 0.688 purity. This is a *demonstration artifact*: at 0.13 linking precision an end-to-end score would conflate linking with summarisation error.

Canonicalisation does not help. Rewriting segments into neutral third-person Hebrew attacks the register confound at the content level rather than the geometric level, and it fails. The rewriting was performed by `gemma-4-31B-it` (4-bit, greedy), one segment at a time, under the editor prompt of Fig. 6: preserve every position, figure, decision and vote, unify register, do not summarise. On 19 protocols it sharpened intrinsic separation (AUC 0.947 → 0.986) but inverted the downstream result (F_1 0.114 → 0.095, precision 0.073 → 0.154), which we first read as the trade our cost model wants. Against gold chains labelled independently of the canonicaliser that reading fails: recall@10 drops 0.948 → 0.862, and the loss concentrates in two-appearance chains. The rewrites compressed to a median 25% of the original length despite that prohibition, dropping the identifiers of Table 4(a).

Automatic segmentation errs in the cheap direction. Table 9 scores the segmenter blind-first: an LLM judge (Claude Fable 5) segments each protocol itself before seeing the system output, and that becomes the reference. The errors are the kind this cost model can absorb. A *merge* — one segment holding two different matters, which files one of them under the wrong entry and cannot be repaired downstream — occurs in only 24 of 356 segments, with 7 of 18 protocols merge-free. A *split* — one matter cut into pieces, each still cleanly about that matter — is common: the system emits $1.58\times$ more segments than the reference. Quality tracks meeting type, near-exact on structured budget sessions (F_1 0.95) and split-heavy on open debate (up to 31 segments against 6). Because splits inflate the stream of arrivals and merged segments have no correct link, every experiment above uses gold spans; the judge is a second opinion, not ground truth.

5 Conclusion

The intuitive failure mode — "Hebrew embeddings are not good enough" — is not what we found: a CPU-side representation separates same- from different-matter segments at AUC 0.954, and retrieval places the right entry in the top 10 every time. What defeats the system is the streaming decision itself, where a 6.9% base rate against a growing candidate set means an excellent false-positive rate still produces more false merges than true ones. The levers that helped were correspondingly structural — judge *distinctiveness* rather than *similarity*, and represent the accumulated matter rather than its first appearance — while larger models in the deciding role did not help at all. On the writing side two readers confirm the journal is coherent and locate the damage where the qualitative analysis put it: the longer a matter recurs, the more each update recaps instead of adding. Our own LLM judge predicts neither reader, never scoring a chain below $60/100$ — including the six they scored $1/5$. An ablation table means little unless its rows share a fitting protocol, and an automatic judge means little until something outside it agrees.

Limitations

Only 36 repeat events: recall at FMR$\leq 5\%$ moves in 2.8-point increments, so large effects are safe and fine rankings are not. Every linking and log-writing result uses gold spans, and our segmenter is scored only against an LLM judge's reference (Table 9), never against human boundaries; all results come from one templated committee. The human assessment is two annotators over 22 chains who agree only moderately on faithfulness (κ_{quad} 0.28) — enough to show the 3B judge tracks neither.

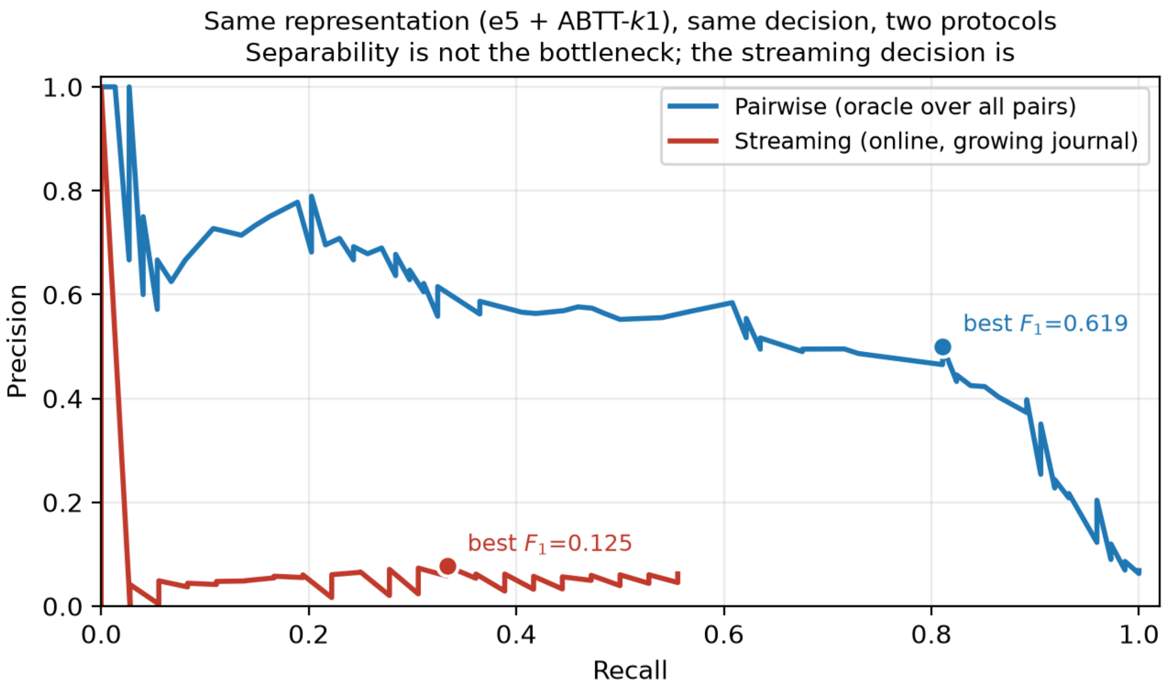

Figure 1: Precision–recall for one representation under both evaluation protocols. The oracle frontier sustains precision near 0.6; the streaming frontier never exceeds ≈ 0.08. The area between them is attributable to the decision setting, not the embedding.

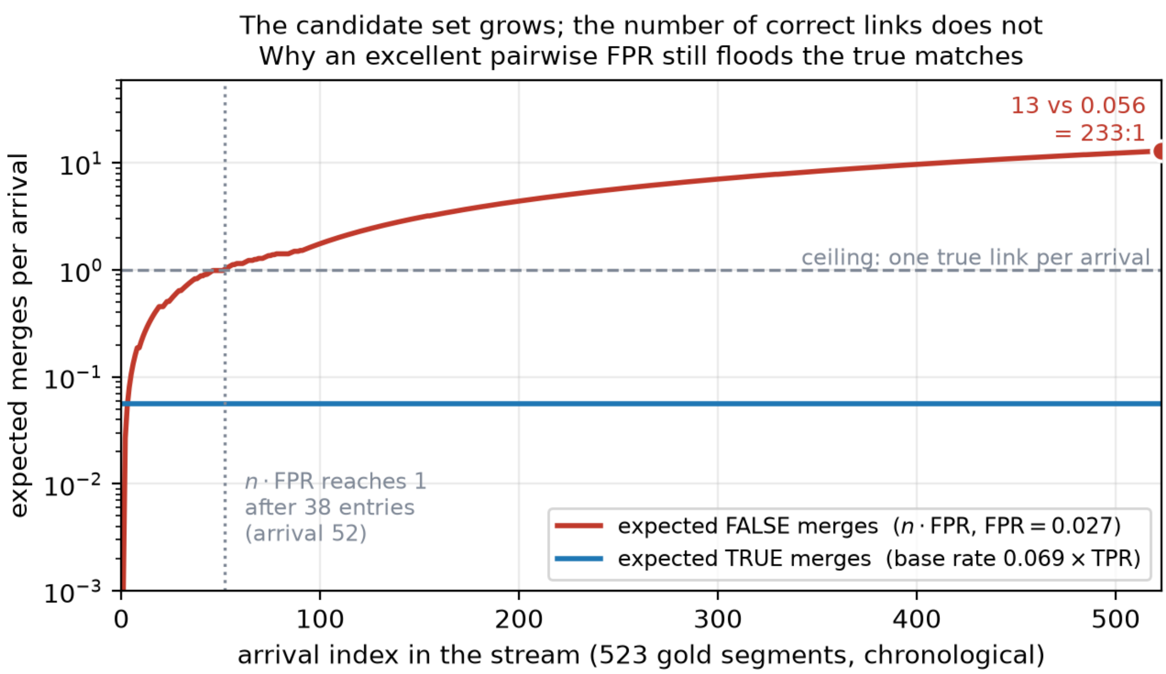

Figure 2: The base-rate arithmetic behind Table 3, over the 523-segment stream. Both curves are counts per arrival, so they share one axis. FPR and TPR are measured over all $\binom{523}{2}$ segment pairs at the threshold that is optimal *pairwise* (θ=0.26); Table 3 reports FMR 0.038 at the same threshold because its denominator is pairs among *recurring* segments only. Because the streaming rule only accepts the top-1 candidate, the red curve counts false-merge *opportunity* rather than realised merges — but the opportunity is what a fixed threshold has to survive, and it grows without bound while the correct answer stays at one link per arrival.

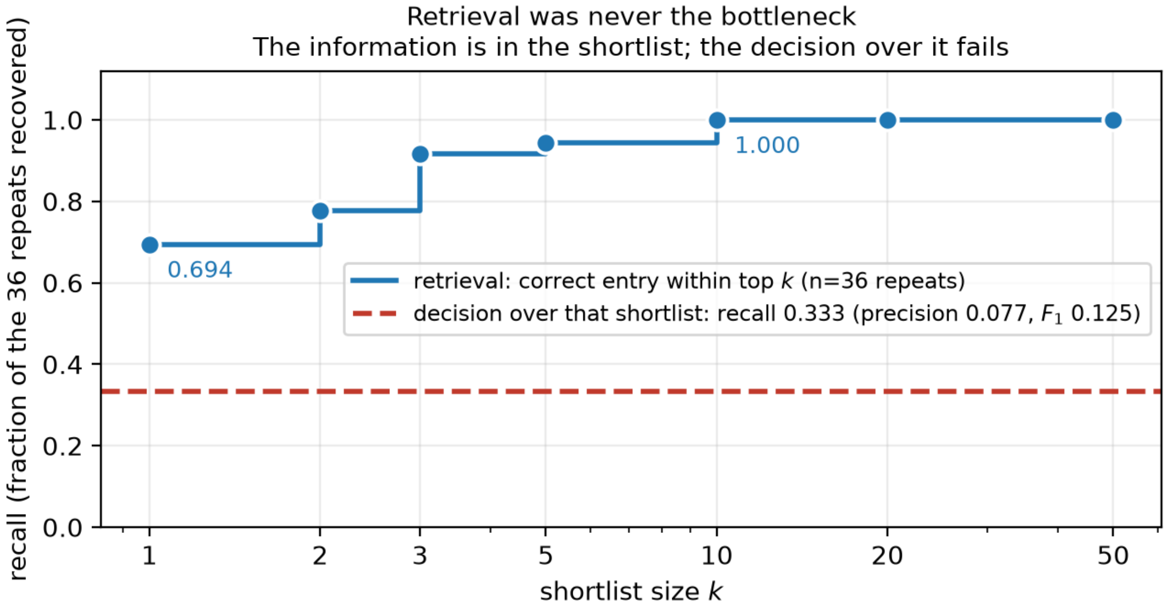

Figure 3: Dense retrieval finds the correct journal entry within the top 10 for every one of the 36 repeat events, and within the top 1 for 69% of them. The decision taken over that shortlist recovers a third of them, at precision 0.077. Both quantities are the same fraction of the same 36 events, so the gap between the curve and the line is the part of the task that scoring, not retrieval, has to solve.

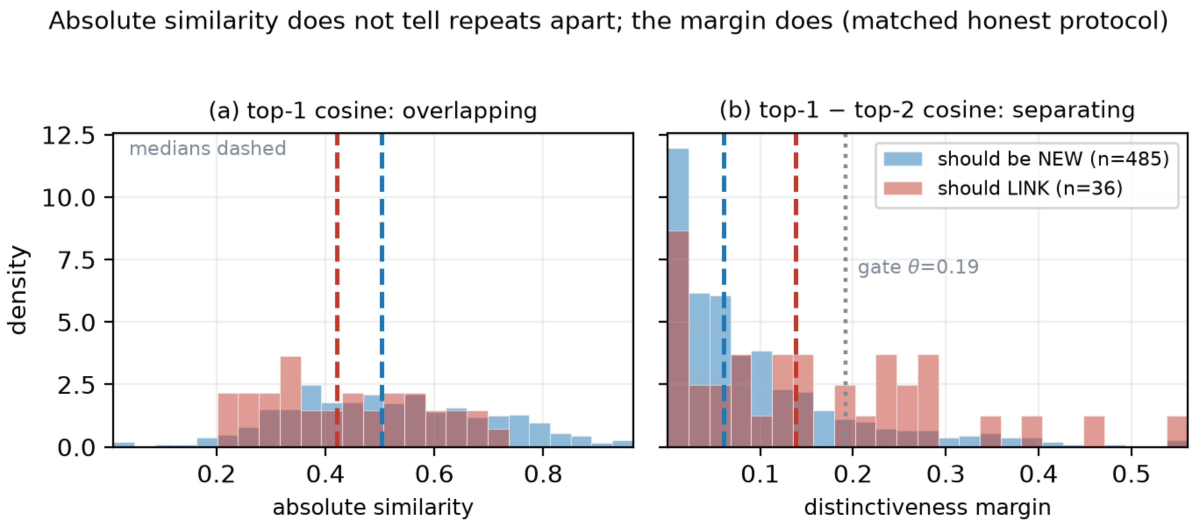

Figure 4: Why Table 5's margin row wins, under that table's matched honest protocol (centering, prefix refit, centroid entries). (a) Absolute top-1 cosine does not identify the arrivals that should link — their median is if anything *lower*, an artifact of shared bureaucratic register across a large journal. (b) The distance to the runner-up does, and the operating gate sits where the two distributions cross. The remaining overlap is why F_1 reaches 0.190 and not more.

```
SYSTEM: You are a precise annotation engine. You always answer with a single
valid JSON object and nothing else - no markdown, no explanations.

You are segmenting the protocol of an Israeli Knesset committee meeting into
discussion segments.

A SEGMENT is a maximal span of consecutive utterances discussing one MATTER
(עניין): a specific bill, budget item/transfer (פנייה / העברה / עודפים),
regulation, appointment, procedure, or policy issue.

RULES:
1. A speaker change is NEVER by itself a segment boundary.
2. A sequence of greetings, congratulations, or thanks about the same occasion
   is ONE segment.
3. A substantive issue raised inside a ceremonial or other speech (a policy
   demand, an announcement of a future hearing, a budget request) STARTS A NEW
   MATTER - even mid-utterance.
4. A VOTE ("הצבעה ... אושרה/נדחתה") ENDS its matter. The next matter begins at
   the FOLLOWING utterance. Never emit a "within" cut on the word הצבעה.
5. Each new budget item announced ("פנייה מס' X", "בקשת העברה", "עודפים",
   "אנחנו עוברים ל...") STARTS a new matter.
6. If unsure whether the matter changed, prefer to cut (over-segmentation is
   recoverable).
```

Figure 5: The per-window segmentation instruction. Rules 1–5 encode committee discourse structure; rule 6 implements the asymmetric cost model.

```
SYSTEM:
אתה עורך לשוני. אתה מקבל מקטע דיון מוועדת הכספים של הכנסת, ובו
דוברים שונים בסגנונות ואוצר מילים שונים. תפקידך לנסח מחדש את תוכן
הדיון בלשון אחידה, ניטרלית ועניינית - לשמר את כל המידע (נושאים,
עמדות, נתונים, החלטות) אך להסיר סגנון אישי, רטוריקה, ומאפייני
דובר. אתה לא מסכם ולא מקצר - אתה מנסח מחדש.

USER:
נסח מחדש את מקטע הדיון הבא בלשון אחידה וניטרלית.

עקרונות:
- שמר את כל התוכן העובדתי: מה נדון, אילו עמדות הוצגו, נתונים
מספריים, החלטות והצבעות.
- אחֵד את הרגיסטר: אותו אוצר מילים ענייני לכל הדוברים, ללא סלנג,
רטוריקה, ברכות או ציטוט סגנוני.
- אל תסכם ואל תקצר באופן אגרסיבי - שמור על אורך דומה לתוכן המהותי
של המקטע.
- כתוב כטקסט רציף בגוף שלישי ("הוצגה עמדה ש...", "סוכם כי...",
"התקיימה הצבעה ש...").
- אל תוסיף פרשנות או מידע שאינו במקטע.

החזר JSON יחיד בלבד:
{"subject": "<כותרת נושא קצרה 3-6 מילים>", "canonical": "<הניסוח
המחדש הניטרלי>",
"decision": "<אושר/נדחה/ללא הצבעה/נדחה להמשך>", "amounts":
["<סכומים או מספרי פניות>"]}
```

Figure 6: The canonicalization prompt (Hebrew). The system role casts the model as a linguistic editor; the instruction requires preserving all factual content (positions, figures, decisions, votes) while unifying register — third person, no slang, rhetoric or speaker style — explicitly forbids summarising or shortening, and requests a single JSON object with subject, canonical text, decision and amounts.

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